MARINDUQUE STATE COLLEGE

DEVELOPMENT OF WEB-BASED TOUR AND TRAVEL MAGAGEMENT SYSTEM FOR GALA TRAVEL AND TOURS

COLLEGE OF ENGINEERING, INFORMATION

AND INDUSTRIAL TECHNOLOGY

BY

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Above all to, Almighty God, who gave strength and intellectual mind to complete this Proposed Project.

- The Developers

DEDICATION

This piece of work is lovingly dedicated to the people who support and inspired us throughout the development of this project.

Our **beloved family**,

Our **friends** and **classmates**,

Our **IIST** and **MSC family**,

To the **Future Developers**,

And above all, to **God Almighty**.

-John Jenoel and Agnes

ABSTRACT

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The Tour and Travel Management System is developed to assist the management of the Gala Travel and Tours in Santa Cruz, Marinduque, in resolving their issues. The main objective of this proposed project is to create a system that automates the manual process of Gala Travel and Tours. The web-based system will enable to minimize the workload, output process and as well as the Tour Management System.

The developers used the modified waterfall technique for the sequential flow of the process of the developing the software. They conducted a feasibility analysis that included technical, operational, and economic feasibility. The availability of resources needed for software development, such as hardware, software, and developer skills, was identified using SWOT analysis in technical feasibility. In terms of operational feasibility, the capabilities and knowledge in handling the developed system, as well as the willingness and capability to support the developed system, were identified. In economic feasibility, the cost of existing operations was gathered to estimate the cost and benefit of the developed system. The developed system's functionality, reliability, usability, efficiency, maintainability, and portability were also evaluated using the ISO 9126 evaluation tool.

The conducted feasibility analysis resulted to satisfaction of the developed system. This system is technically feasible due to the developers' capability in developing the system. Moreover, it is operationally feasible due to the user's capability in using and handling the developed system as well as the willingness to implement it. Furthermore, it is economically feasible where the organization can meet its return on investment in one (1) year, five (5) and three (3) weeks. Lastly, the conducted evaluation obtained the grand mean average of 4.548 with the descriptive ratings of Very Good. Thus, the system requirements met the needs of the organization and also the user of the system.

Further, the Agency Management system is created to make it simple to navigate the agency and complete tasks efficiently. Also, it is created to reduce the number of problems that can occur when using an online system. This could save a significant amount of time and effort.

Keyword: ISO9126, Web Reservation System, Travel Agency

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CHAPTER 1

INTRODUCTION

Background of the Project

Following the COVID-19 epidemic, business travel is once again anticipated, and many organizations have learned that their outdated systems may not be suited for managing business travel at scale. Most organizations rely on labor-intensive manual processes that waste time and money rather than spending money on technology that can automate many of these tasks and streamline every step of the process.

With the rapid advancement in internet-based technology, it has been a great experience for most of the travelers in a country or around the world. Online booking is an online platform that use internet-based technology that helps travelers to book their desired destination. A company that carries out the following tasks is referred to as a travel agency. These tasks include quoting prices, making reservations, booking travel itineraries and lodging, organizing travel insurance, foreign exchange, and travel-related paperwork.

On the other hand, an Online travel agency, so called OTA, is an automated intermediary focusing in giving consumer services, products, and reservation capability on World Wide Web (Law et al., 2004, p. 100) [1].

Manual process in business is not really the deal in this era. This kind of process is outdated, you won't get any cost benefit, all the money you spend in the process can be used to a system or process which will benefit any organization. That is why we are proposing to use a Web-based booking system for the organization GaLa Travel and Tours. This system will truly boost their entire process and will attract more tourist or travelers to book their desired destinations through online.

Project Objectives

General Objectives

The general objective of this project is to create a system that automate the manual process of GaLa Travel and Tours. The web-based system will enable to minimize the workload, output process and as well as the Tour Management System.

Specific Objectives

This project specifically aimed to:

1. To further analyze the process of the existing system.
2. Design and create a web system that will help GaLa Travel and Tours in their online booking system.
   - 2.1 Create a system where client can book or avail a package tour to their desire destination.
   - 2.2 Create a system where it allows the admin to change the tour package details altogether.
   - 2.3 Create a system that allows payment online using Gcash.
3. Asses the performance of the proposed system based on functionality, reliability, usability, efficiency, portability and maintainability.
4. Implement the system.

Significance of the Project

The proposed system is aiming to replace the old process used by GaLa Travel and Tours and to enhance their booking process and the entire travel management system. The said system would be a great help to the following.

To the Manager. The admin of the system can easily manage or track the important transactions. From updating all the tour packages package rate, and booking calendar.

To the Tourist. They can easily book a tour package to their desired travel destination online.

To the Developers. This system project can be a steppingstone in sharpening skills, expanding knowledge about software design and development. They may also get to understand and discover something in this project that they can use in their future works in web development.

To the Future Developers. The future developers can be benefited by this project system, particularly when they want to use it as their reference and guide when creating another system that is most likely to be related with the project. This project system may become their quickest guide and reference use when the future developers want to develop another Tour Management System.

Scope and Limitations of the Project

The proposed project is focused on creating a web-based system that will help GaLa Travel and Tours with their Travel Management. The Travel Management system will be giving a boost with the organizations process in terms of booking, viewing and updating tour packages, payment and the record monitoring.

The admin has access to all the system capabilities. The admin would have to log in first. The system enables the admin to change or update tour package information like, transportation, rate, date available, and location. Customers who want to avail tour package will have to register on the website. They can only view the tour details it includes the date, transportation, rate, and the location. Can book or avail a package. They cannot update tour details.

Definition of Terms

For a better understanding of the discussions of this project, the following terms are defined both conceptually and operationally:

Online Travel Agency- is an intermediary agent selling travel products and services such as the airlines, car rental, cruise lines, Hotels Accommodation, railways, and vacation packages on behalf of suppliers using internet as a medium.

Tour Management System- Tour management system is used to book trips, manage upcoming trips, and track and report on previous trips.

Web-based - is software that is connected to a network or internet.

Web-based application - refers to an application that is designed to run the web-browser.

CHAPTER 2

REVIEW OF RELATED LITERATURES AND STUDIES

This chapter presents the review of related literatures and studies underlying the framework of the project. It includes the conceptual model of the project.

Travel Agency

Travel agent acts as an intermediary serving between the various service providers and travel consumers. They sell and market the products and services provided by the suppliers such as hotel, transport operators, airlines, railways, cruise etc. Travel agents serve as a link between the travel suppliers and the tourists. "A travel agency may be an individual, a business firm, or company which acts as an intermediary in the sales and promotion of different travel related services selling to its client and earns commission" [1].

"A travel agency is defined as a business that performs the following functions: quote fares, rates, make reservations, arrange travel tickets and accommodation, arrange travel insurance, foreign currency, documents and accept payments. European commission, define travel agency as a retailer, leisure, and business tourists, selling flights (charter or scheduled), hotels, car rentals, foreign exchange, travel coverage and other facilities. The travel agency acts as an intermediary, accessibly linking clienteles with the supplier of travel products [2].

India is a huge country in terms of area and economy; with the growth of per capita income the people are becoming more passionate for travel and simultaneous increasing the demands of travel agency. Thomas Cook, SOTC, Cox and Kings, Yatra.com, make my trip are some of the iconic names in the travel world [1].

Online Travel Agents (OTAs)

Increasing numbers of FITs are turning to online travel agents (OTAs), companies that aggregate accommodations and transportation options and allow users to choose one or many components of their trip based on price or other incentives [3].

Examples of OTAs include Booking.com, Expedia.ca, Hotwire.com, and Kayak.com. OTAs are gaining popularity with the travelling public; in 2012, they reported online sales of almost $100 billion and almost triple that figure, upward of $278 billion, in 2013 [4].

In early 2015 Expedia purchased Travelocity for $280 million, merging two of the world's largest travel websites. Expedia became the owner of Hotels.com, Hotwire, Egencia, and Travelocity brands, facing its major competition from Priceline [5].

In addition, Although OTAs can provide lower-cost travel options to travelers and the freedom to plan and reserve when they choose, they have posed challenges for the tourism industry and travel services infrastructure. As evidenced by the merger of Expedia and Travelocity, majority of popular OTA sites are owned by just a few companies, causing some concern over lack of competition between brands. Additionally, many OTAs charge accommodation providers and operators a commission to be listed in their inventory system. Commission-based services, as applied by Kayak, Expedia, Hotwire, Hotels.com, and others, can have an impact on smaller operators who cannot afford to pay commissions for multiple online inventories [4].

Finally, governments are stepping in as they see OTAs as a barrier to collecting full tax revenues on accommodations and transportations sold in their jurisdictions. OTAs frequently

charge taxes on the retail price of the component; however, they purchase these products at a discount, remitting only the portion collected on the lesser amount to the government. In other words, the OTA pockets the difference between taxes collected and taxes remitted [6].

The Development of Online Billing and Invoice Management System for Saudi Small Companies

Invoicing and billing software for various business needs primarily assists service providers and freelancers in managing, sending professional invoices online, and tracking their status. In general, all small businesses in Saudi Arabia face a variety of issues when it comes to managing and tracking customer invoice status, which is primarily due to these businesses' failure to adapt to new technology [7].

Tracking Online Reservation History with the Billing

The Hudson Group Knowledge Base was designed for clients of The Hudson Group and their suite of transportation management applications. The Knowledge Base provides many quick self-help answers to common or frequently asked questions. Using the online Billing Utility, Hudson clients who accept reservations from the Web Reservations System, can view the history of reservations that are recorded online for a single day, week, month, or other period of time. A common use of the billing utility is to obtain statistical data for online reservation system, useful when reviewing monthly statements from the Hudson group reflecting totals for reservation recorded online [8].

Marsman Drysdale Travel INC. (MDTI)

MDTI is one of the best travel and tours Company in the Philippines. MDTI was established in 1975 due to lack of tourist industry here in the Philippines. Located in the Makati Business District as their head office, MDTI operates all over the country. MDTI are served to give quality services and products to the customers. Keeping their moral, professional, ethical standards in their operations, making great leadership in the business travel industry and honesty as they could meet the customer needs. The system will allow clients to look for the available services of the company with the use of online reservation. MDTI reservation system is similar to the proposed system of the proponents in terms of minimizing data redundancy, inconsistency on data in reservation, and reducing paper works. Nowadays internet and computer have big parts in their daily lives because it makes their work faster. The clients could also save their time and effort by booking online (www.mdti.com.ph).

Travel Booking System

A travel booking engine is web-based software. It allows consumers and travel agents to book flights, hotels, holiday packages, cars, and other services online. This is a much - needed software for the travel and tourism industry, as it become a dynamic tool for online travel business [9].

Conceptual Model of the Project

On the basis of the foregoing concepts theories and findings of related literatures, studies and insights taken from them, a conceptual model was developed as shown below.

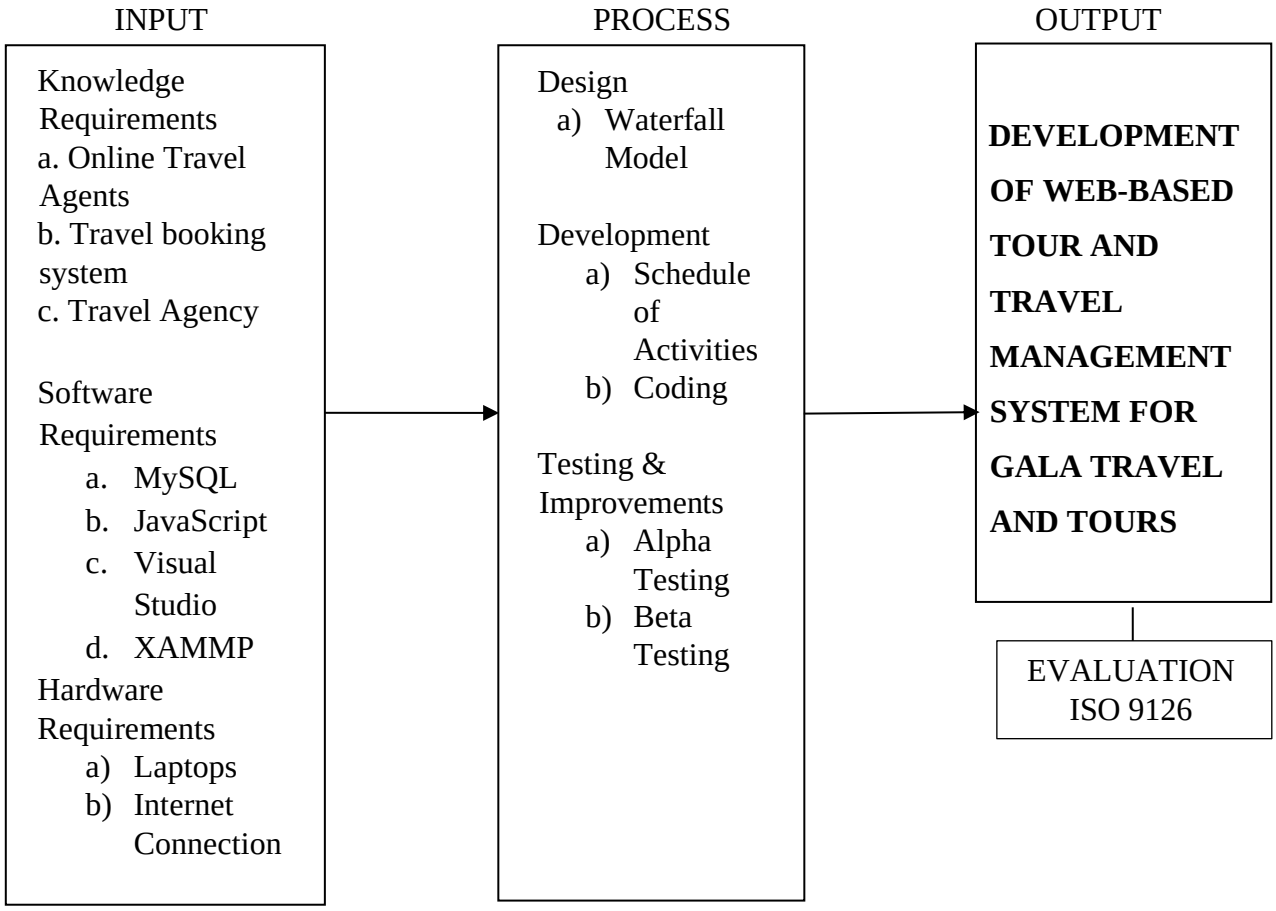

Figure 1. Conceptual Model of Development of Web-Based Tour and Travel Management System for Gala Travel and Tours

Figure 1 shows the different stages of the processes. The input stage consists of three sub-components: knowledge requirements, software requirements, and hardware.

The next stage is the Process; it contains the design, development, testing and improvements for the developed system.

These are the inputs and processes that result in the Development of Web-Based Tour and Travel Management System for Gala's Travel and Tours.

Chapter 3

Methodology

This section presents the Conceptual Model of the study or project, System Architecture, Requirement Analysis, Feasibility Analysis Procedure which includes technical, operational and economic feasibility.

Requirements Analysis

This section describes the steps involved in gathering the data needed to create the developed system.

An interview with the management of Gala's Travel and tours was conducted in order to gather information for an evaluation of the current manual process, which will be used to design the system.

In addition, the developers gathered information about the Travel and Tours Process through observation and survey questionnaires. This strategy's goal is to identify problems with the manual system as well as potential solutions for creating the developed system.

Description of Waterfall Model

Software development is done using the waterfall paradigm, which consists of several phases. It is easy to go from one stage of this software development process to the next. The name of the model refers to how the flow resembles a waterfall. It is often referred to as the conventional waterfall model.

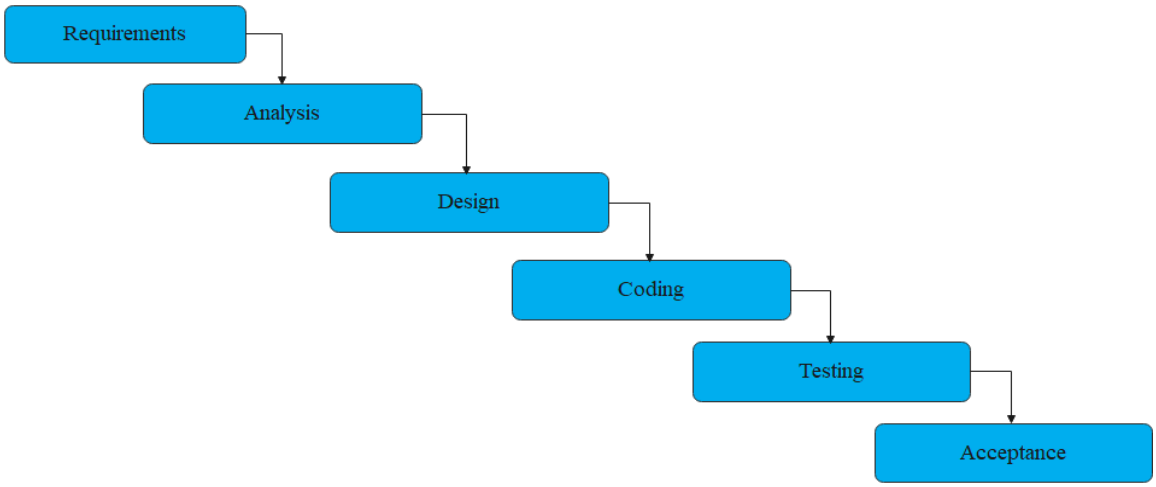

Figure 2. ***Waterfall Model***

Source: pngegg.com

Figure 2. Illustrates the software development Process. This will be the model that will use to create the system. In Waterfall Model, each phase must be completed before the next phase can begin. The entire software development process is divided into phases. In a sequential order, the output of one phase becomes the input for the next phase. This means that any phase of the development process can start only after the previous one has finished.

Requirements

The first phase of the waterfall model is the requirements, were we gather information. All the information that gathered will be used for creating a system.

Analysis

The process in developing a new system is to identify all potential system needs and document them in a requirement specification document. The information and requirements that are gathered in from the client will be analyzed for the validity of the creating a system.

Design

The System Design is followed in the waterfall model. Were the developers used an architectural design for the created system. Also, the flow and the process of the created system were presented as well as the input and output processes are analyze in the system.

Coding

In Coding, it has different tools that are uses in creating a system. Its goal is to translate the concept of the system into code using a programming language.

Testing

After the coding, testing Phases are followed were the developers were going to test their system. They used alpha testing to identify if the system is working properly and have an error or issues.

Acceptance

The final phase of the waterfall model is acceptance, at which time the system was allowed to develop its own operating methodology. Based on the provided documentation and the results of the implementation aid, there was a plan for how the system will be maintained.

Feasibility Analysis Procedure

With the data and information provided by the organizations Manager and Staffs, the developers can gain a better understanding of the current services, strength, weaknesses, opportunities, and threats of the travel organization. The data and information gathered will be used to develop and implement a system as its main objective.

Technical Feasibility

With the use of technical feasibility, the organization's hardware and software resources were determined. The developers scrutinized the attributes and capabilities of the Agency to check if the hardware, software, and networking tools fulfilled the system's basic hardware and software requirements. Additionally, the developers examined whether the new system will address the issues the organization is facing and also they examined the usability of the system in processing transaction and the user's capacity to operate it. Several factors were considered upon conducting the technological feasibility project, this includes whether the proposed system had enough potential for future need, whether the required resources and the required resources were available within the organization and whether the proposed system is required.

Strengths

- Enhanced personal and writing skills.
- Develops confidence
- Increases thinking capacity
- Promotes Interaction

Weaknesses

- Developers lack experience in system development.
- Time - The lack of time prevented developers from finishing all of the resources needed by the system.

Opportunities

- Increased understanding
- An environment where one can practice critical thinking
- Enhance your writing skills
- Following the collection of all the data, the project research to boost self-assurance was operationally feasible.

Threat

• Plagiarism — developers should double-check the sources of their work, particularly if it is not their own ideas or just borrowed resources.

Operational Feasibility

To gauge their readiness to take part in the project's planning and development, the developers interviewed the Agency's management and staff as part of the operational feasibility assessment. For instance, the supporters asked system users about their level of technical expertise. The developers concluded that the project study was operationally feasible after gathering all of the data since the respondents provided all of the necessary and relevant information for the creation of the system.

Table 1. *SWOT Analysis of GaLa Tours and Travels*

STRENGTHS	WEAKNESSES
● Provide the best travel deals ● Informing users about the price changes ● Affordable rates	● New in the market, more marketing are needed ● Staffs are lacking in technological skills ● Lack of technological implementation
OPPORTUNITIES	**THREATS**
● Current services are not personalized enough ● Growth in Tourism ● Collaboration with local businesses	● Not enough funds for strong marketing campaigns ● Strong competition

The proposed system is feasible, according to the developers SWOT Analysis. One of the organization's strengths is that they provide the best deals and affordable tour rates. On the other hand, one of the organization's flaws is its lack of technological implementation, as well as its staff's lack of technological knowledge.

Economic Feasibility

In order to determine the Agency's economic viability, the developers conducted an interview about their transactions with their tour packages. All of the collected data was used by the developers to conduct cost-benefit analyses, return on investment analyses, and payback time analyses to determine whether the current system is economically viable. All of the costs incurred by the system, as well as all of the benefits expected upon its operation, are illustrated in the secure sheet presentation. While ROI determines the amount spent after the system is put in place. Furthermore, the payback period determines the number of years take to recover the development costs. Because the developed system reduces the agency's operational expenses in terms of transaction costs, it was financially feasible. The formula used by the developers were as follows:

$$\text{Breakeven Ratio} = \frac{\text{Cost Benefits} - \text{Cumulative Benefits}}{\text{Cost Benefits}}$$

Breakeven Point = Breakeven Year + Breakeven Ratio

Chapter 4

RESULTS AND DISCUSSION

The description of the existing system, the requirements specification, results of the feasibility analysis, the implementation result, and the testing result are all presented in this Chapter.

Description of the Existing System

An interview was conducted with the Gala Travel and Tours, according to the manager, the Gala Travel and Tours Agency has been using a manual process for recording the traveler or the tourist who wants to visit the different tourist destination in Marinduque.

The manager is responsible for compiling reports and making reservations for all the travelers. Before making a reservation, guests should call or message the management on Facebook. The manager requests complete information from the customer or guests, including full name, age, address, and phone number. The manager also inquires as to when will they visit and how many days they will stay. Once the guests have arrived, the payment is made to the manager.

Requirements Specification Functional Requirements

1. Make Booking Reservation
   - Guests are able to make booking reservations through online
2. View Different Tour Packages
   - Guests and Admin can view different tour packages
3. Issue Bills

- Admin can issue bills and invoice

4. Manage Guest

- Admin are able to manage the record of every guest

6. Manage Reports

- Admin can manage reports like list of cancelation reports and sales reports

Non-Functional Requirements

1. Usability

- The system should be easy to use and operate

2. Reliability

- The system should not encounter errors or bugs

3. Performance

- The system should respond to the user's request

4. Security

- The system should require username and password in order to access the system.

Result of the Feasibility Analysis

Technical Feasibility

The availability of the resources and equipment used, as well as the developer's capacity to develop the system, were used to determine technical feasibility.

SWOT Analysis of Technical Capabilities of the Developers

Strengths

- The developers are enrolled in an IT education course.
- The developers have the right available resources to develop the system
- The developers have technical skill in writing and developing a system.

Weaknesses

- Power Interruption which may cause delay and loss of data.
- Slow internet connection.
- The developers are often busy with other things and responsibilities.
- The developers are not yet expert in programming.

Opportunities

- The developers are familiar with the process of creating a system.
- The developers will be able to complete the project on schedule.
- The developers continuously worked in the development due to the availability of resources and skill in system development.

Threats

- Manuscripts is delayed in submitting
- Delay in creating the system

Operational Feasibility

Table 1. *Computer System Components and Specification*

Gala Travel and Tours	
System Components (Hardware and Software)	Specification
Desktop Computer Processor	Intel Core i3
RAM	4 GB
Hard Disk Capacity	512GB
Operating System	Windows 11
Printer	EPSON L3210

The table 1 shows the ICT Resources that the Gala Travel and Tours uses throughout their processes. Each resources is sufficient enough for the developed system.

Economic Feasibility

The economic feasibility shows the tables in the existing system. Table 2 shows the supplies and materials used in existing system. Table 3 shows the project increased of supplies 10% per year and Table 4 shows the cost of existing system.

Table 2. *Supplies and Materials Used in the Existing System*

Description	Quantity	Unit	Unit Cost	Total Cost
Record Book	10	Piece	₱ 140.00	₱ 1,400.00
Short Bond Paper	10	Ream	₱ 250.00	₱ 2500.00
Folder Short	40	Piece	₱ 10.00	₱ 400.00
Ballpen Ordinary	2	Box	₱ 80.00	₱ 160.00
Pencil	1	Box	₱ 60.00	₱ 60.00
Puncher	1	Pieces	₱ 200.00	₱ 200.00
Filling Cabinet	2	Pieces	₱ 7,000.00	₱ 14,000.00
Computer/Laptop	1	Unit	₱ 21,000	₱ 21,000.00
Paper Fastener Plastic	6	Box	₱ 20.00	₱ 120.00
Total				**₱ 39,840.00**

Table 2 shows the materials used in the existing system. The total cost of the supplies and materials in the existing system is Php39,840.00.

Table 3. *Project Increase of Supplies 10% per Year*

Description	Year 1	Year 2	Year 3	Year 4	Year 5
Record Book	₱ 1,400.00	₱ 1,540.00	₱ 1,694.00	₱ 1,863.40	₱ 2,049.74
Short Bond Paper	₱ 2500.00	₱ 2,750.00	₱ 3,025.00	₱ 3,327.50	₱ 3660.25
Folder Short	₱ 400.00	₱ 440.00	₱ 484.00	₱ 532.4	₱ 585.64
Ballpen Ordinary	₱ 160.00	₱ 176.00	₱193.6	₱ 212.96	₱ 234.26
Pencil	₱ 60.00	₱ 66.00	₱ 72.6	₱ 79.86	₱ 87.85
Puncher	₱ 200.00	₱ 220.00	₱ 242.00	₱ 266.2	₱ 292.82
Filling Cabinet	₱ 14,000.00	------	-------	-----	-------
Computer/Laptop	₱ 21,000.00	---------	--------	-------	-----
Paper Fastener Plastic	₱ 120.00	₱ 132.00	₱ 145.2	₱ 159.72	₱ 175.69
Total	**₱ 39,840.00**	**₱ 5324.00**	**₱ 5856.4**	**₱ 6442.04**	**₱ 7,086.25**

The table 3 shows the project price increased of some of the materials used in the existing system. The price increased valued to 10% per year. Other materials that could record in another year do not have price increase.

Table 4. *Cost of Existing System*

Description	*Total*
Total cost of supplies and materials	₱ 39,840.00
Energy Consumption(400 per *12months)	₱ 4,800.00
Other expenses (Transportation, Photocopy, allowance)	₱ 5,000.00
Total Cost	**₱ 49,640.00**

Table 4 shows the total cost of the system, the total cost of supplies and materials, and the total cost energy consumption in one year. Budgets for the maintenance of the computer and others are also allotted.

Table 5. *Operational Cost of the Existing System*

	Year 1	Year 2	Year 3	Year 4	Year 5
Total cost of supplies and materials	₱ 39,840.00	₱ 5324.00	₱ 5856.4	₱ 6442.04	₱ 7,086.25
Energy Consumption	₱ 4,800.00	₱ 5,280.00	₱ 5,808.00	₱ 6,388.8	₱ 7,027.68
Other Expenses	₱ 5,000.00	₱ 5,500.00	₱ 6,050.00	₱ 6,655.00	₱ 7,320.5
Total Cost	**₱ 49,640.00**	**₱ 16,104.00**	**₱ 17,714.00**	**₱ 19485.00**	**₱ 21,434.43**

Table 5 shows the operational cost being used by the existing system. The supplies and materials, energy consumption, and other expenses are indicated. Since the existing system uses paper based execution of processes, a huge number of supplies and materials are acquired.

Table 6. Developmental Cost

Description	*Amount*
Energy Consumption	₱ 1,500.00
Transportation	₱ 2,000
Internet Fee	₱ 4,000
Documentations	₱ 5,000
Total Cost	₱ 12,500

Table 6 shows the Developmental cost of the developed system. The energy consumption, transportation, internet fee and the documentations fee.

Table 7. *Supplies and Materials Used in the Developed System*

Description	Quantity	Unit	Unit Cost	Total Cost
Short Bond Paper	2	Ream	₱ 250.00	₱ 500.00
Folder Short	10	Piece	₱ 10.00	₱ 100.00
Ballpen Ordinary	2	Box	₱ 80.00	₱ 160.00
Pencil	1	Box	₱ 60.00	₱ 60.00
Puncher	1	Pieces	₱ 200.00	₱ 200.00
Printer Ink	2	Bottle	₱ 160.00	₱ 320.00
Paper Fastener	1	Box	₱ 40.00	₱ 40.00
Stapler	1	Piece	₱ 50.00	₱ 50.00
Total				**₱ 1,430.00**

Table 7 shows the supplies and materials that are used in the developed system. These supplies and materials are needed in creating and printing needed reports using the system.

Table 8. *Projected Increase of Supplies 10% per Year*

Description	Year 1	Year 2	Year 3	Year 4	Year 5
Short Bond Paper	₱ 500.00	₱ 550.00	₱ 605.00	₱ 665.5	₱ 732.05
Folder Short	₱ 100.00	₱ 110.00	₱ 121.00	₱ 133.1	₱ 146.41
Ballpen Ordinary	₱ 160.00	₱ 176.00	₱ 193.6	₱ 212.96	₱ 234.26
Pencil	₱ 60.00	₱ 66.00	₱ 72.6	₱ 79.86	₱ 87.85
Puncher	200.00	-----	--------	-----	-------
Printer Ink	320.00	₱ 352.00	₱ 387.2	₱ 425.92	₱ 468.51
Paper Fastener	40.00	₱ 44.00	₱ 48.4	₱ 53.24	₱ 58.56
Stapler	50.00	-----	-----	-----	-----
Total	₱ 1,430.00	₱ 1,298.00	₱ 1,427.8	₱ 1,570.58	₱1,727.64

Table 8 shows the projected price increase of some of the materials used in the existing system. The price increase valued up to 10% per year. Other materials that can be reused for another year do not have price increase.

Table 9. *System Operational Cost of the Developed System*

Description	Total
Total cost of supplies and materials	₱ 1,430.00
Developmental Cost	₱ 12,500.00
Total Cost	**₱ 13,930.00**

Table 9 shows the operational cost being used by the developed system in one year. It includes the supplies and materials used in developed system which cost Php1,430.00, and the developmental cost of Php12,500.00 with a total of Php13,930.00.

Table 10. *Operational Cost of the Developed System*

	Year 1	Year 2	Year 3	Year 4	Year 5
Total cost of supplies and materials	₱ 1,430.00	₱ 1,298.00	₱ 1,427.8	₱ 1,570.58	₱ 1,727.64
Developmental cost	₱ 12,500.00	₱ 13,750.00	₱ 15,125.00	₱ 16,637.5	₱ 18,301.25
Total Cost	₱ 13,930.00	₱ 15,048.00	₱ 16,552.8	₱ 18,208.08	₱ 20,028.89

Table 10 shows the operational cost being used by the developed system in projected 5 years. The supplies and materials, and the developmental cost (energy consumption, internet fee, transportation, and for documentation).

Table 11. *Cost and Benefit Analysis*

Description	Amount					
	Year 0	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Cost of the Existing System		₱49,640.00	₱16,104.00	₱17,714.00	₱19,485.00	₱21,434.43
Cost of the Developed System		₱13,930.00	₱15,048.00	₱ 16,552.8	₱18,208.08	₱20,028.89
Developmental Cost	₱12,500.00					
Cost Benefit	₱12,500.00	35,710.00	(1,479.5)	(1,161.2)	(1,276.9)	(1,405.54)
Cumulative Benefit	₱12,500.00	23,210.00	21,730.5	20,569.3	19,292.4	17,886.86
Net Benefit	**₱17,886.86**					
Break Even Ratio	**0.53**					
Break Even Year	**Year 1**					
Payback Year	**1.53**					

Table 11 shows the cost and benefit analysis of the system. To determine the cost and benefit analysis of the system, the cost of the developed system is subtracted from the existing system.

To compute the breakeven ratio, formulas are as follows:

Breakeven Ratio = $\frac{Cost\ Benefit - Cummulative\ Benefit}{Cost\ Benefit}$

$= \frac{49{,}640.00 - 23{,}210.00}{49{,}640.00}$

Breakeven Ratio = 0.53

Breakeven Year = Year 1

Breakeven Point= Breakeven Year + Breakeven Ratio

= 1 + 0.53

Breakeven Point= 1.53

In the feasibility analysis that was conducted, the project could receive the benefit of the system after one (1) year, five (5) months and three (3) weeks.

Design of the Developed System

Description of the Developed System

The description of the developed system is based on the Gala Travel and Tours Agency Information System's process, where it was developed to provide the quickest and easiest means of booking and compiling reports in the agency.

The Agency Management Information System is a web-based system that can only be accessed by visitors who have access to the internet. The Guest can book and pick a tour packages. Before picking a tour packages, the guest must register an account. The admin can view the number of users registered in the website. Where the admin can change the tour package details and manage the booking.

Testing Result

Functions	**Test data**	**Expected data**	**Actual result**	**Test after debugging**	**Remarks**
User Login					
-Autorized	Valid username and password	The system will log in the authorized user.	The authorized user had successfully logged in.	The authorized user has successfully logged in.	The user can successfully log in.
- Unautorized	Invalid/incor rect username and password	An unauthorized user was unable to access the system.	An unauthorized user was unable to successfully log in.	An unauthorized user was unable to log in.	Unauthorized users are unable to log in.
Booking	To make a booking, the guest must fill out the form.	Guest will make bookings.	The guest had made a successful booking reservation.	Booking can be made through online.	Guests can successfully book accommodations online

Evaluation Result

Table 12. *Summary of Evaluation Results of Beta Testing*

Criteria	Mean	Descriptive Rating
Functionality	4.6	Very Good
Reliability	4.533	Very Good
Usability	4.520	Very Good
Efficiency	4.613	Very Good
Maintainability	4.563	Very Good
Portability	4.463	Very Good
Grand Mean	**4.548**	**Very Good**

Table 12 shows the summary result of evaluation using the ISO 9126 Evaluation Instrument. The grand mean average was 4.548 which were interpreted as “Very Good”. This means that the developed system met the requirements and processes of the organization. (See Appendix E)

Test Result of ISO 9126

The project's developers used ISO 9126 as an evaluation tool to assess the functionality of the developed system and for determining system and system functionality the user's satisfaction.

Based on the preceding findings and project objectives, the following conclusions were reached:

- In terms of functionality, the developed system can interact with the system environment, which means that all functions have been implemented and are fully functional. The system also has a feature that prevents unauthorized users from evaluating it.
- In terms of reliability, the software encounters no errors or bugs during the system evaluation. And when a failure occurs, the software can also be restored to normal operation and does not terminate.
- In terms of usability, the system displays simple instructions makes the system simple to use and operate.

CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusion and recommendation which are based on the developed system.

Summary of Findings

The project aimed to develop Agency Management Information System which would help the organization to minimize the effort and work in managing the Agency's bookings and reports.

As a result of analyzing the feasibility, the developed system would be economically feasible within five (5) years, helping to lower operational expenses in terms of the transaction cost of the Gala Travel and Tours Agency. The hotel would have achieved a breakeven ratio of 0.53 and a breakeven point in one (1) year, five (5) months, and three (3) weeks if the developed system had been used. The process of determining which hardware and software requirements the agency can meet is referred to as "technical feasibility." The practical feasibility was possible because the agency management could provide all the necessary and important information for the development of the system.

Conclusions

The developers conclude that implementing the designed system will yield significant benefits. The Agency Management Information System was created with the goal of making it easier to navigate the hotel and to provide quick operations within the process. The established system met the project's objectives. The company could save a significant amount of time and effort this way.

Recommendations

According to the findings of the proposed project, it is recommended to browse the page on a computer or laptop to ensure the system's ability to run effectively. To avoid any problems with the booking, the developers also recommend having a good internet connection. Before continuing to the next level of the system procedure, the system user should read the system alert and prompt messages.

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APPENDICES

Appendix A
Letter to Organization

September 27, 2022

MR. JOHN JENOEL P. QUIMORA
MS. AGNES R. CASTILLO
Owner
Gala Travel and Tours
Agency
Santa Cruz Marinduque
Subject: *Request a letter to conduct an Interview*

Dear Mr. Quimora and Ms. Castillo,

Good Day!

The undersigned is a 3rd year of Bachelor of Science in Information Systems students of Marinduque State College (MSC) who are currently working on a proposed study entitled **"DEVELOPMENT OF WEB-BASED TOUR AND TRAVEL MAGAGEMENT SYSTEM FOR GALA TRAVEL AND TOURS".**

Relative there to form, we humbly request your adjuration as my Client.

With your experience and expertise we are confident that are proposed system will improved handily.

We are hoping for your favorable response should have any related concern. Please email us at galatravelandtours@gmail.com or contact us at 09505501631.

Thank you very much!

Respectfully yours,

John Jenoel Quimora

Agnes Castillo
Proponents
Information System Students

Noted by:

ENGR. ALVIN P. REGENCIA
Project Adviser

Appendix B
Interview Questions

1. What is the existing system of the organization?

2. What are the services or operations being offered and done in your organization?

3. What are the common problems encountered by your organization?

4. What are the processes and services of your organization?

5. Where do you usually store or keep your files?

6. What are the causes and effects of your problems?

7. How do you prepare your report?

8. What are the inconveniences in generating them?

Appendix C
WBS (WORK BREAKDOWN STRUCTURE)

1. Initiating Process

1.1 Form a Project Team

1.2 Identify the Target Organization

1.3 Prepare a letter to the Organization

1.4 Conduct Interview

1.5 Gathering Data

1.6 Prepare Topic Presentation

1.7 Present Topic Presentation

2. Planning

2.1 Identifying the existing problem of the Organization

2.2 Identifying the scope and limitation

2.3 Determining the Significance of the Project

2.4 Gathering Related Literatures and studies

2.5 Determining the Tools and Techniques to be used

2.6 Identifying the Software and Hardware Requirements

2.7 Feasibility Study

2.8 Describing the Existing System

2.9 Work Breakdown Structure

3. **Executing Process**

    3.1 Design User Interface

    3.2 Coding

    3.3 System Testing

    3.4 Installation on Client's Site

4. **Monitoring and Controlling Process**

    4.1 Completing Documentation

    4.2 Determining syntax and logical errors

5. **Implementation**

6. **Closing**

    6.1 Formalized turnover of the developed system

APPENDIX D

EVALUATION INSTRUMENT FORM

Name: _____
Category of Evaluation: _____Users

INSTRUCTIONS: Please evaluate the prototype by using the given scale and placing a checked marked under the corresponding numerical rating**.**

Numerical Rating	**Equivalent**
5	Excellent
4	Very Good
3	Good
2	Fair
1	Poor

INDICATORS	**5**	**4**	**3**	**2**	**1**
FUNCTIONALITY					
1. The funcions included in the system are fully functional (accurateness).					
2. The software has available all the functions required for its execute (adequacy)					
3. The software has provision to prevent unauthorized users from accessing the system (security)					
4. The software has secure access through password (secure access)					
RELIABILITY					
1. The software encounters no error or bug (maturity)					
2. The software does not terminate despite errors or bugs are encountered (fault tolerance)					
3. The software can restore its normal operation when failure occurs (recoverability)					
USABILITY					
1. It is to understand the concept and application (intelligibility)					
2. It is easy to perform its function (intelligibility)					

3. It is easy to learn how to use (learnability)					
4. The software facilities the users entry (learnability)					
5. It is easy to operate and control (operability)					
EFFICIENCY					
1. The software's response time is appropriate (time)					
2. The software's execution time is appropriate (time)					
3. The software uses determines the error or bug encountered during operation (analysability)					
4. The resources used are appropriate (resources)					
PORTABILITY					
1. It easy to adapt to other environment (adaptability)					
2. It easy to installed in other environment (capability to be installed)					
3. It is in agreement with portability standard (conformity)					
4. It is easy to use to replace another program (capability to replace)					
MAINTAINABILITY					
1. Analysability (Software deficiencies or causes of failures to be identofied).					
2. Changeability (to enable specified madifications to be implemented).					
3. Stability (to minimize the impact of software changes on the system).					
4. Testability (the ability to validate software that has been modified).					

APPENDIX E

EVALUATION RESULT

1		ISO 9126 TABULATED RESULT																											GRAND MEAN
2		WEB-BASED TOUR MANAGEMENT																											
3		Functionality				Reliability			Usability					Effeciency				Maintainability				Portability							
4		1	2	3	4	1	2	3	1	2	3	4	5	1	2	3	4	1	2	3	4	1	2	3	4				
5	Respondent 1	5	5	4	5	4	4	4	4	5	4	5	4	5	4	4	5	4	4	5	5	3	4	4	4				
6	Respondent 2	5	5	4	4	4	5	4	4	5	5	4	4	5	4	5	4	5	4	5	5	4	4	4	4				
7	Respondent 3	5	4	5	4	5	5	5	4	5	5	5	5	4	5	5	5	5	4	5	5	5	5	5	5				
8	Respondent 4	5	4	4	5	4	5	4	5	5	4	4	5	5	5	4	5	4	5	4	4	4	5	5	5				
9	Respondent 5	5	4	5	4	4	5	5	4	4	4	5	5	4	5	5	4	4	5	5	5	5	5	4	4				
10	Respondent 6	5	4	5	5	5	4	4	4	5	4	4	5	5	5	4	5	5	4	4	5	4	4	4	5				
11	Respondent 7	4	4	5	4	4	5	5	4	4	5	5	4	4	4	4	5	5	4	5	5	5	5	5	4				
12	Respondent 8	5	4	5	5	5	5	4	5	5	5	4	5	4	5	4	4	4	5	4	4	5	4	5	5				
13	Respondent 9	4	5	4	4	4	5	5	4	4	5	5	4	5	4	4	5	5	4	3	5	4	5	5	4				
14	Respondent 10	4	5	4	5	5	4	4	5	4	5	4	5	5	5	5	4	5	5	5	4	5	4	5	4				
15	Respondent 11	5	5	5	4	4	5	4	4	5	4	5	4	4	4	5	5	4	5	5	5	3	4	5	4				
16	Respondent 12	4	5	5	5	4	4	5	5	4	5	4	5	5	5	5	5	5	4	5	4	5	5	5	5				
17	Respondent 13	5	5	4	4	5	5	4	4	5	5	5	5	4	5	5	4	5	5	4	5	5	5	4	4				
18	Respondent 14	4	5	5	5	5	4	5	4	4	4	4	4	5	4	5	4	4	4	5	4	4	4	4	5				
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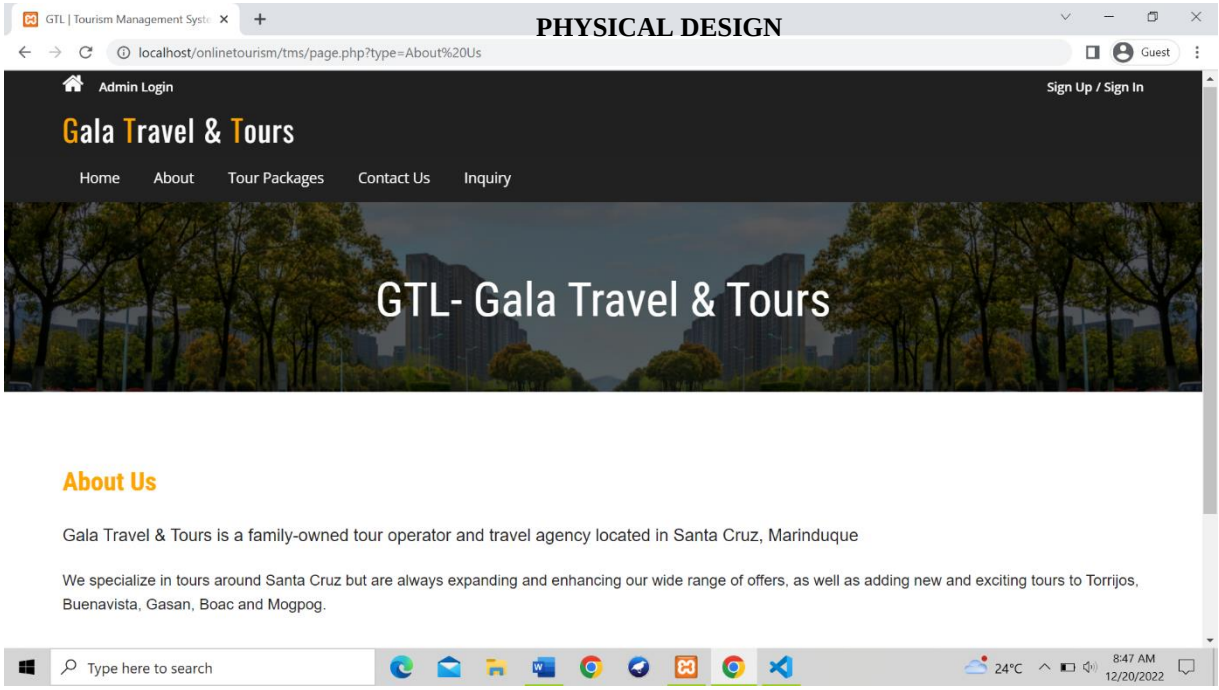
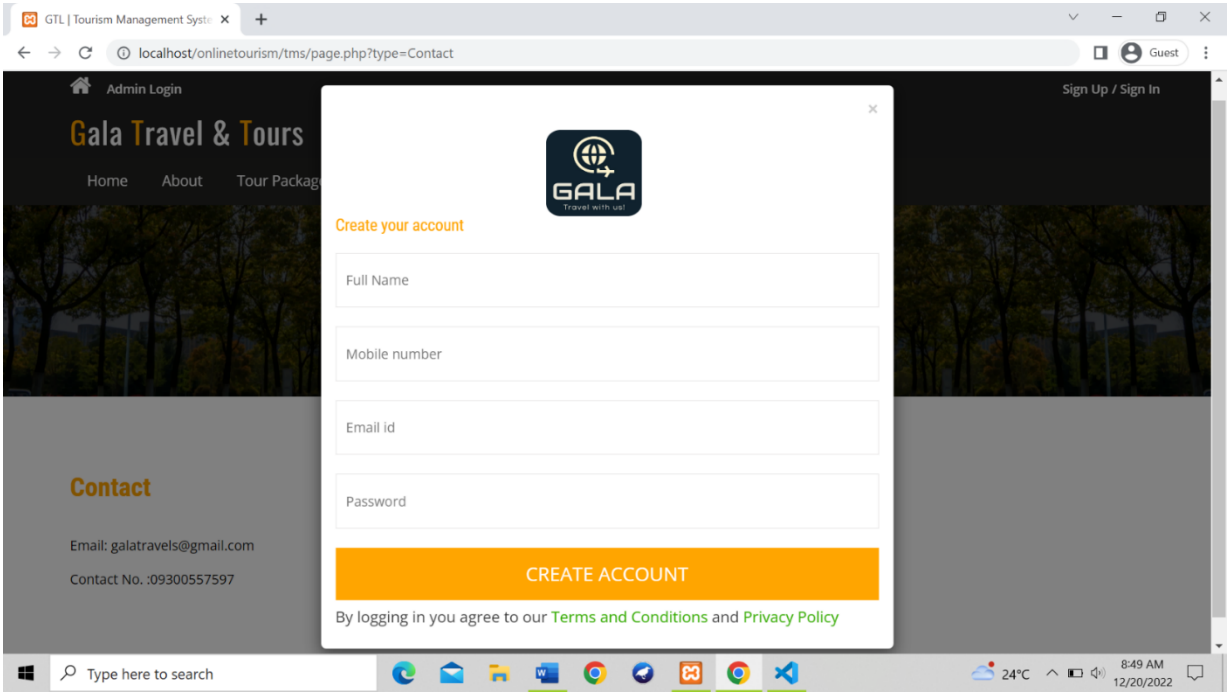
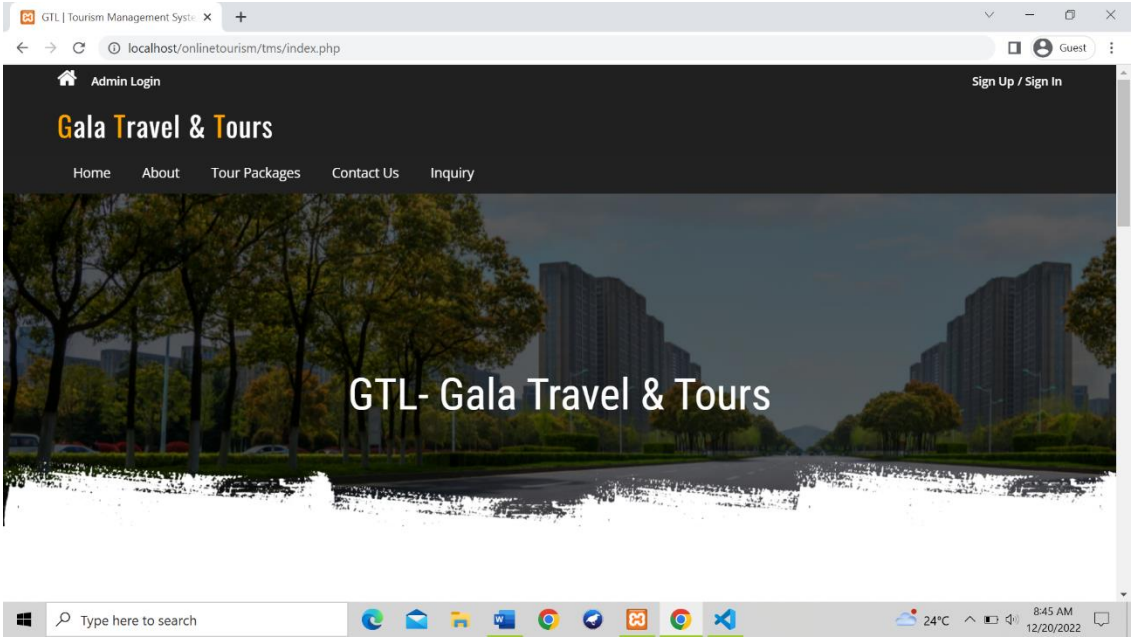
20	Respondent 16	4	5	4	5	4	4	4	4	5	4	5	4	5	4	5	5	4	5	3	5	5	5	4	4				
21	Respondent 17	4	5	5	4	5	4	5	5	4	5	4	5	5	5	5	4	5	4	5	4	5	4	4	5				
22	Respondent 18	4	5	5	5	5	5	4	4	5	4	5	4	5	4	5	5	5	5	5	5	5	4	5	4				
23	Respondent 19	5	5	5	4	5	5	4	5	4	5	5	5	5	5	5	5	4	5	5	4	4	4	5	4				
24	Respondent 20	5	5	5	5	4	5	5	4	5	4	5	5	5	4	5	4	5	4	5	4	5	5	5	4				
25																													
26																													
27		4.600	4.700	4.600	4.500	4.500	4.650	4.450	4.300	4.550	4.550	4.600	4.600	4.650	4.550	4.650	4.600	4.600	4.500	4.550	4.600	4.450	4.450	4.600	4.350				
28	WM				4.600			4.533					4.520				4.613				4.563				4.463				
29																													
30																													4.548

APPENDIX F

PHYSICAL DESIGN






GTL | Tourism Management Syste
localhost/onlinetourism/tms/index.php
Admin Login
Sign Up / Sign In
Gala Travel & Tours
Home
About
Tour Packages
Contact Us
Inquiry
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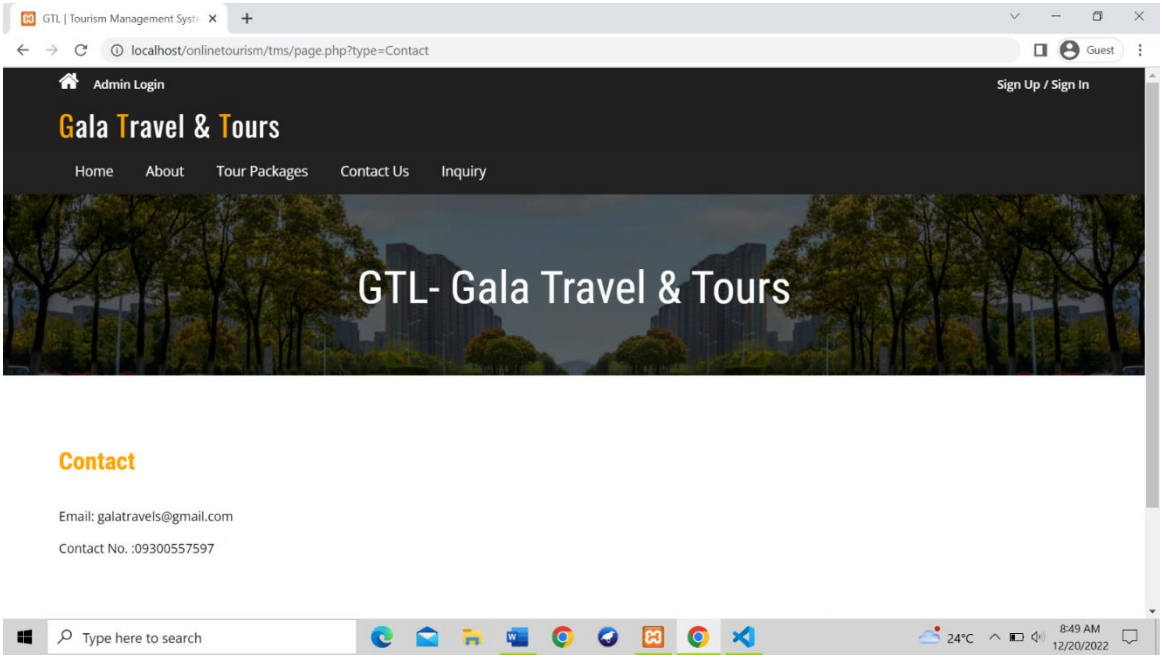
Type here to search
24°C
8:45 AM
12/20/2022


GTL | Tourism Management Syste
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Admin Login
Sign Up / Sign In
Gala Travel & Tours
Home
About
Tour Packages
Contact Us
Inquiry
GTL- Gala Travel & Tours
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Email: galatravels@gmail.com
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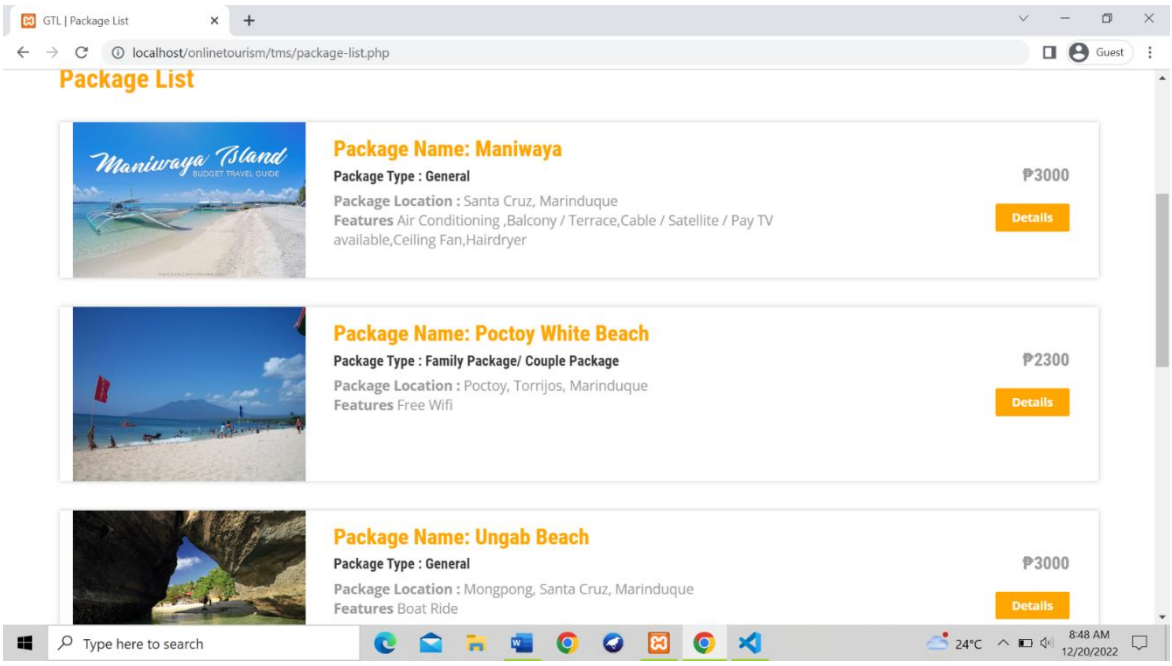
12/20/2022


GTL | Package List
localhost/onlinetourism/tms/package-list.php
Package List
Package Name: Maniwaya
Package Type : General
₱3000
Package Location : Santa Cruz, Marinduque
Features Air Conditioning ,Balcony / Terrace,Cable / Satellite / Pay TV available,Ceiling Fan,Hairdryer
Details
Package Name: Poctoy White Beach
Package Type : Family Package/ Couple Package
₱2300
Package Location : Poctoy, Torrijos, Marinduque
Features Free Wifi
Details
Package Name: Ungab Beach
Package Type : General
₱3000
Package Location : Mongpong, Santa Cruz, Marinduque
Features Boat Ride
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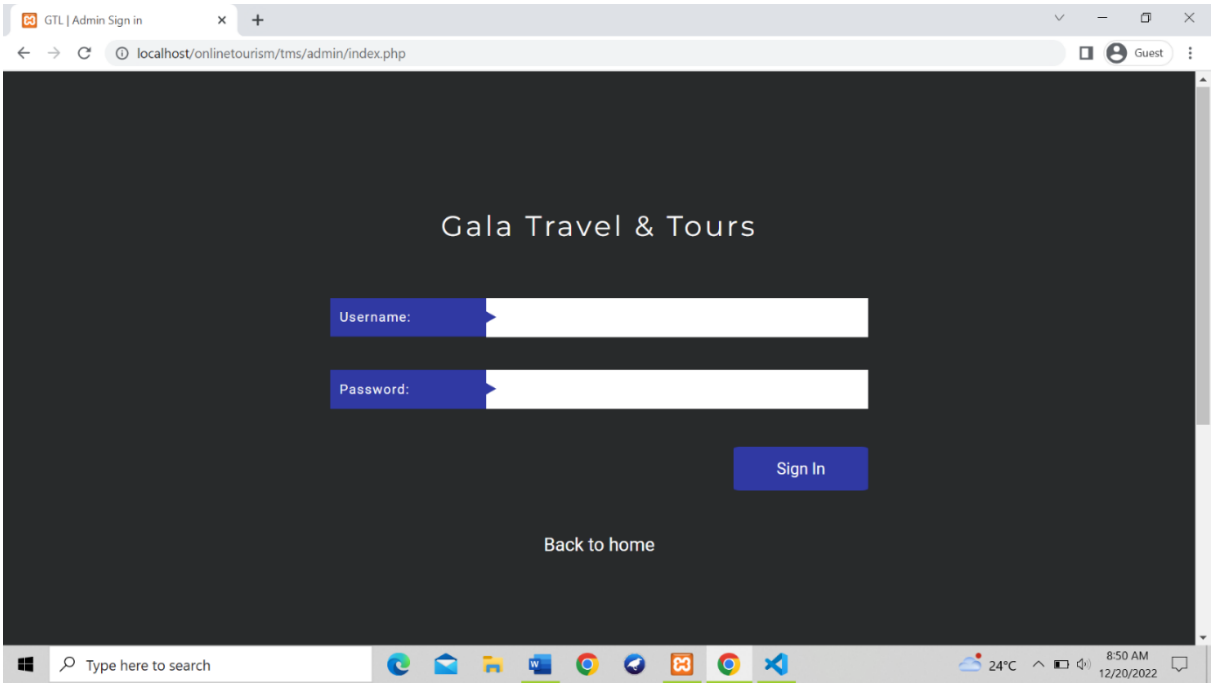
12/20/2022


GTL | Admin Sign in
localhost/onlinetourism/tms/admin/index.php
Gala Travel & Tours
Username:
Password:
Sign In
Back to home
Type here to search
24°C
8:50 AM
12/20/2022

CURRICULUM VITAE

DEVELOPERS BACKGROUND

AGNES R. CASTILLO

Address: San Isidro Santa Cruz Marinduque

Email: castillorioflorido0302@gmail.com

Contact Number: 09756186353

Personal Information

Age:	**20**
Gender:	**Female**
Father's Name:	**Agapito Castillo**
Mother's Name:	**Myrna Rioflorido**
Civil Status:	**Single**
Religion:	**Catholic**

Educational Background

College:	College of Engineering, Information and Industrial Technology Undergraduate
	Bachelor of Information System
	Marinduque State College
	Matalaba, Santa Cruz Marinduque
Senior High School:	General Academic Strand
	Botilao National High School
	Botilao Santa Cruz Marinduque
Junior High School:	Botilao National High School
	Botilao Santa Cruz Marinduque
Elementary:	San Isidro Elementary School
	San Isidro Santa Cruz Marinduque

DEVELOPERS BACKGROUND

JOHN JENOEL QUIMORA

Address: Biga, Santa Cruz, Marinduque

Email: quimorajenoel@gmail.com

Contact Number: 09487993001

Personal Information

Age:	**20**
Gender:	**Male**
Father's Name:	**Crestituto Quimora**
Mother's Name:	**Marita Quimora**
Civil Status:	**Single**
Religion:	**Catholic**

Educational Background

College:	College of Engineering, Information and Industrial Technology Undergraduate
	Bachelor of Information System
	Marinduque State College
	Matalaba, Santa Cruz Marinduque
Senior High School:	Pre-Baccalaureate Maritime Specialization
	Educational Systems Technological Institute
	Murallon, Boac, Marinduque
Junior High School:	Alobo National High School
	Alobo Santa Cruz Marinduque
Elementary:	Biga Elementary School
	Biga Santa Cruz Marinduque